

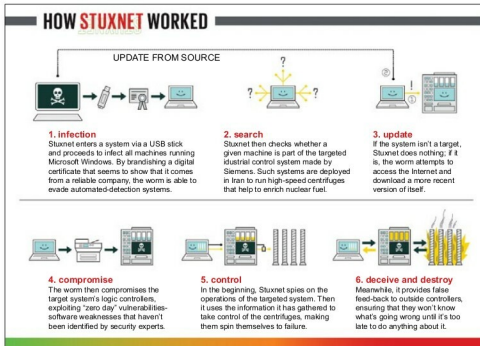


Figure 3: How Stuxnet worked

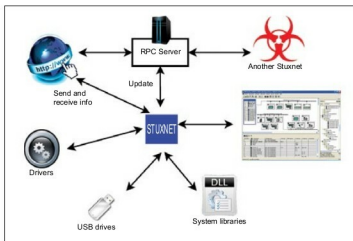


Figure 4: Components of Stuxnet

malicious software. These range from modifying system libraries, targeting the Step 7 SCADA controller, the use of a command and control server (via RPC), installation of malicious driver code in system software like drivers, etc. Stuxnet is also capable of downloading auto updates, which update an older version installed in a system. It recursively communicates to the command and control server covertly, to pass on information about its spread, the devices affected and possible ways of getting updated. Stuxnet auto updates via a built-in peer-to-peer network. This kind of auto updating leads to different versions and different attack vectors. The recent version of Stuxnet uses 'LNK vulnerabilities', whereas the older version uses *Autorun.inf*.

LNK vulnerability: Stuxnet registers the code in an infected Windows computer. Whenever USB drives are inserted, Stuxnet automatically copies the code to the drive. A very interesting fact is that the code will work to infect any of three machines. If the drive is already infected, a new version

of Stuxnet is updated to the drive for further infection. An LNK vulnerability contains four files with the *.lnk* extension in addition to Stuxnet DLL, which are used to execute the payload in different versions of Windows.

Autorun.inf: *Autorun.inf* is an auto run file which automatically runs the code in USB drives whenever the drive is inserted in a Windows machine. Here, the command and code for infection is inserted and configured in the *autorun.inf* file. Hence, the Windows OS ignores the Stuxnet data portion in the *autorun.inf* file.

WinCC

WinCC is an interface to SCADA systems; Stuxnet searches for WinCC and connects to the database via a hard-coded password. It then attacks the

database using SQL commands by uploading and replicating a copy of code on a WinCC computer.

Network shares

Stuxnet can use Windows' shared folders to propagate itself in a LAN. It drops and schedules a dropper file to execute it on remote computers.

Print spooler vulnerability (MS10-061)

Stuxnet copies and shares itself to remote computers using this vulnerability and then executes itself to infect the remote computers. In general, Stuxnet replicates into two, and then it copies itself into the *C:\Windows\system* folder with zero day privilege escalation and infects the computer.

SMB vulnerability (MS08-067)

Server message block (SMB) is a network file-sharing protocol. Stuxnet modifies the path and sends data through a malformed path between remote computers.

Step7 project vulnerability

Stuxnet infects the executable of WinCC Simatic manager and modifies the shared libraries, DLLs, libraries for shared objects, code and data. Then it copies the additional code into the *step7* directory, and connects to the command and control server for remote execution of payloads.

The attack phase

Stuxnet consists of two modules—the user module and kernel module.

User module: In the user module, there are several functions to do the following operations. First, Stuxnet itself injects malware code into the actual running process. This results in the execution of the malware code in the target address